

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	ISE549
Subject Title	Management of Innovation and Technology
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	None
Objectives	1. To develop the understanding of the range, scope, and complexity of the innovation systems and the technology incorporated (i.e. organizational innovation system, industrial, and regional etc) 2. To develop insights on the issues and challenges in managing innovation (systems) in the innovation process 3. To identify the critical skills required for managing the innovation process 4. To offer some practice in defining and working out strategic management problems related to technological innovation and corporate entrepreneurship.
Intended Learning Outcomes	Upon completion of the subject, students will be able: <i>Category A relates to: Professional/Academic knowledge and skills, and Category B: Attributes for all-roundedness.</i> a. To be able to identify practical innovation system/practices for companies (i.e. organizational, and regional) b. To be able to perform situational analysis of company's innovation strategies c. To formulate strategies applicable to high-technology venture capital projects d. To be able to implement and manage the innovation projects as a team

Subject Synopsis/ Indicative Syllabus	1. <u>Importance of Technology and Innovation in Economic and Social Development</u> Risks and rewards of technological innovation; role of government policy in promoting technology and innovation; role of firm-level strategy in the global market 2. <u>Innovation Types and factors affecting</u> Understanding on factors affecting innovation adoption; Diverse types of innovations; the heterogeneous impact of diverse innovations on firm performance 3. <u>Organizational Learning as an approach to drive innovations</u> Understanding how firms apply knowledge management and learning to maintain their advantage, why some firms are faster than others in creating new innovations and responding to change. 4. <u>Importance of Technological Evolution</u> Lessons regarding the evolution of technology; the main concept of competition changes in accordance with the technological evolution 5. <u>Technology Adoption</u> Understanding on differences between early vs. late adoption and its performance heterogeneity 6. <u>Firm Boundary Decisions</u> Understanding on firm boundary; the impact of firm boundary decisions on performance; key reasons that firms change their boundaries; relationship between firm boundary decision and firm capability; relationship between firm boundary decision and industry structure																														
Teaching/Learning Methodology	This subject is offered via weekly classes. A mixture of lectures, tutorial exercises, and case studies will be used to deliver the various topics in this subject. Some of the topics will be covered in a problem-based format where this enhances the learning objectives. Case discussion and project activities will take place against a background of conceptual materials that are acquired through selected readings and brief lectures pertaining to the theme of each session.																														
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr><tr><td>1. Case Analysis (Individual)</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>2. Project proposal (Group)</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d			1. Case Analysis (Individual)	20%	✓	✓	✓	✓			2. Project proposal (Group)	10%	✓	✓	✓			
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																											
		a	b	c	d																										
1. Case Analysis (Individual)	20%	✓	✓	✓	✓																										
2. Project proposal (Group)	10%	✓	✓	✓																											

	3. Test	30%	✓	✓	✓	✓		
	4. Online discussions	10%	✓	✓	✓	✓		
	5. Project Report (Group)	20%	✓	✓	✓			
	6. Project presentations	10%	✓	✓	✓	✓		
	Total	100 %						
	Assessment comprises of an assignment, a group project with individual and group components, contributions in class and in online discussions.							
Student Study Effort Required	Class contact:							
	▪ Lectures							15 Hrs.
	▪ Seminar/Case Studies							24 Hrs.
	Other student study effort:							
	▪ Preparation for seminar/case studies							28 Hrs.
	▪ Preparation for project presentation and report writing							70 Hrs.
	Total student study effort							120 Hrs.
Reading List and References	1. Melissa A. Schilling, Strategic Management of Technological Innovation, 7th edition, McGraw Hill, New York, 2022. 2. Burgelman, Robert, Clayton Christensen and Steven Wheelwright, Strategic Management of Technology and Innovation, McGraw-Hill/Irwin (ISBN: 0-07-2536950), latest edition. 3. Tushman, Michael and Philip Anderson, Managing Strategic Innovation and Change, Oxford Press, latest edition. 4. Allen, Thomas J., Managing the Flow of Technology, MIT Press, Cambridge, Massachusetts, latest edition More readings will be provided during the course delivery.							

Classes are held every **Tuesday from 6:30pm to 9:20pm in room Z512** unless otherwise stated.

ISE 549 Teaching Plan

Week	Topic	Pedagogy	Assessment	Instructor(s)
1 (1 Sept 2026)	Subject orientation Lecture: Knowledge and Innovation in the age of digitalisation, basic concepts in Strategies and Business Management	Lecture Discussions		Andy Sun & Eric Tsui
2 (8 Sept 2026)	Lecture: Creativity and Innovation, The “WOW” effect, Innovation processes and systems, Types of Innovations	Lecture Discussions		Eric Tsui
3 (15 Sept 2026)	Lecture: Knowledge and Innovation, Innovator’s dilemma and factors affecting innovation success Assignment – Case Analysis	Lecture Discussions	Case analysis released	Eric Tsui
4 (22 Sept 2026)	Lecture: Corporate Entrepreneurship, Intrapreneurs Group project: Explanation of the project	Lecture Discussions Tutorial	Group project announced	Eric Tsui
5 (29 Sept 2026)	Lecture: Importance of Technological Evolution	Lecture Discussions		Andy Sun
6 (6 Oct 2026)	Lecture: Role of government policy, Strategies for high tech venture capitalist projects	Lecture Discussions	Case analysis due	Andy Sun
7 (13 Oct 2026)	Lecture: Firm and its environment, Business Model Canvas, Business Model Innovation, Firm boundaries	Lecture Discussions	Project proposal due	Eric Tsui
8 (20 Oct 2026)	Strategies for Technology Innovation companies	Discussions Presentations		Eric Tsui
9 (27 Oct 2026)	Lecture: New Product Development	Lecture Discussions		Andy Sun
10 (3 Nov 2026)	Lecture: Critical skills for managing innovation process and Learning Organisations	Lecture Discussions		Andy Sun
11 (10 Nov 2026)	Guest Lecture	Lecture Discussions		Andy Sun
12 (17 Nov 2026)	Test			Eric Tsui
13 (24 Nov 2026)	Project Presentations	Presentation s	Project Presentation	Andy Sun & Eric Tsui
	Submission of Final report		Final report due	